

Ex. No: 6 Semaphore Implementation

C PROGRAM

```
#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

#include <semaphore.h>

#include <sys/mman.h>

#include <sys/wait.h>

int main()

{

sem_t *sem;

sem = mmap(NULL, sizeof(sem_t),

PROT_READ | PROT_WRITE,

MAP_SHARED | MAP_ANONYMOUS,

-1, 0);

sem_init(sem, 1, 1);

if(fork() == 0)

{

sem_wait(sem);

printf("Child Process Entering Critical Section\n");

sleep(3);

printf("Child Process Leaving Critical Section\n");

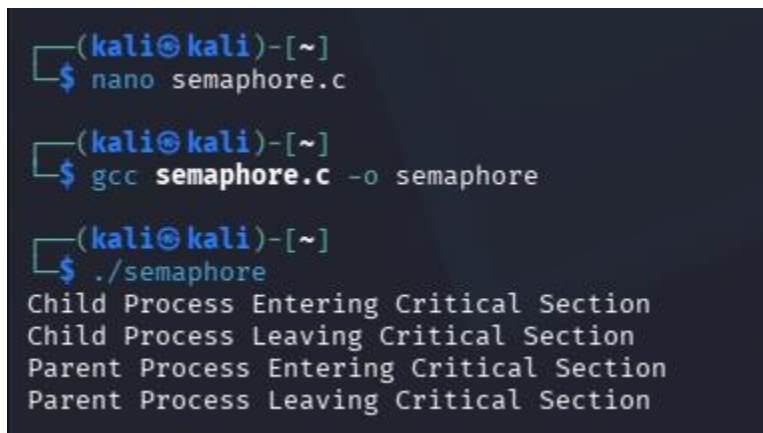
sem_post(sem);

exit(0);

}
```

```
sem_wait(sem);  
printf("Parent Process Entering Critical Section\n");  
sleep(3);  
printf("Parent Process Leaving Critical Section\n");  
sem_post(sem);  
wait(NULL);  
sem_destroy(sem);  
return 0;  
}
```

Output:

A terminal window with a dark background and light blue text. The prompt is (kali@kali)-[~]. The user enters 'nano semaphore.c'. The prompt changes to (kali@kali)-[~]\$. The user enters 'gcc semaphore.c -o semaphore'. The prompt changes to (kali@kali)-[~]\$. The user enters './semaphore'. The output shows four lines: 'Child Process Entering Critical Section', 'Child Process Leaving Critical Section', 'Parent Process Entering Critical Section', and 'Parent Process Leaving Critical Section'.

```
(kali@kali)-[~]  
$ nano semaphore.c  
  
(kali@kali)-[~]  
$ gcc semaphore.c -o semaphore  
  
(kali@kali)-[~]  
$ ./semaphore  
Child Process Entering Critical Section  
Child Process Leaving Critical Section  
Parent Process Entering Critical Section  
Parent Process Leaving Critical Section
```

Shell Script:

```
#!/bin/bash  
  
LOCKFILE="/tmp/mylock"  
  
while [ -f "$LOCKFILE" ]  
  
do  
  
sleep 1  
  
done
```

```
touch "$LOCKFILE"
```

```
echo "Entering Critical Section"
```

```
sleep 5
```

```
echo "Leaving Critical Section"
```

```
rm -f "$LOCKFILE"
```

Output:

```
(kali㉿kali)-[~]  
$ nano lab-6.sh  
  
(kali㉿kali)-[~]  
$ chmod +x lab-6.sh  
  
(kali㉿kali)-[~]  
$ bash lab-6.sh  
Entering Critical Section  
5  
6  
Leaving Critical Section  
  
(kali㉿kali)-[~]  
$ 5  
5: command not found  
7  
  
(kali㉿kali)-[~]  
$ 6  
6: command not found
```